

Exploratory Data Analysis (EDA)

Understanding Your Data Before Modeling

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- Exploratory Data Analysis (EDA) is the process of analyzing datasets to summarize their main characteristics.
- It often involves visual methods and descriptive statistics.
- **Goal:** Gain insights, detect patterns, spot anomalies, and test assumptions.

EDA importance

- Understand the structure of the data.
- Detect missing values, outliers, or errors.
- Discover relationships between variables.
- Choose the right modeling approach.
- Guide feature engineering.

Types of Data to Explore

- **Numerical Variables:** e.g., height, income.
- **Categorical Variables:** e.g., gender, country.
- **Time-Series Data**
- **Text or Image Data**

Each type requires different techniques for exploration.

- **Central tendency:** Mean, Median, Mode
- **Dispersion:** Variance, Standard Deviation, Range
- **Distribution shape:** Skewness, Kurtosis
- **Tip:** Use `df.describe()` in Pandas to summarize data.

Visualizing Data – Univariate Analysis

- **Histograms:** visualize distributions
- **Boxplots:** detect outliers
- **Bar charts:** show category frequencies

Helps in understanding single-variable behavior.

Visualizing Data – Multivariate Analysis

- **Scatter plots:** for correlations
- **Heatmaps:** of correlation matrices
- **Grouped boxplots:** compare distributions across categories

Helps in finding relationships between variables.

Dealing with Missing Data & Outliers

- Detect missing values using `isnull()`
- Handle by:
 - Removing rows
 - Imputing (mean, median, model-based)
- Detect outliers using:
 - Boxplots, z-scores, or IQR method

Common Pitfalls to Avoid

- Ignoring data types (e.g., treating categorical as numerical)
- Overlooking missing data
- Relying too much on automatic plots without interpretation
- Jumping to modeling without EDA

Conclusion & Tools for EDA

- EDA is a critical first step in data science.
- It shapes your understanding and decisions.
- **Tools:**
 - Python (Pandas, Seaborn, Matplotlib)
 - R (ggplot2, dplyr)
 - Notebooks (Jupyter, RMarkdown)

“Look at your data before you model it.”